

Mathematical induction

Def. Let $A(k)$ be a statement for $k \in \mathbb{N}$. Mathematical induction is a proof method used to show that a given statement (for example A) is true for all natural numbers* (i.e. $A(n)$ holds for $\forall n \in \mathbb{N}$). It consists of 2 steps:

1. **Base Case:** Prove that the statement is true for the first number(s)* (for example show that $A(1)$ holds)

2. **Induction step:** Here we need to show that for every number $n \in \mathbb{N}$, that if the statement is true for n , then it is also true for $n+1$ (i.e. if $A(n)$ holds, then $A(n+1)$ also holds).

In order to do that, we assume that the statement holds for some arbitrary $k \in \mathbb{N}$ (**Induction Hypothesis**) and using that assumption we prove that the statement also holds for $k+1$.

* There are cases where the statement we are trying to prove doesn't hold for the natural numbers smaller than some number $m \in \mathbb{N}$. In those cases we can still use induction to prove that the statement holds for all natural numbers $\geq m$. In such cases we need to be careful so that in the base case we prove the statement for m and not for numbers smaller than m .

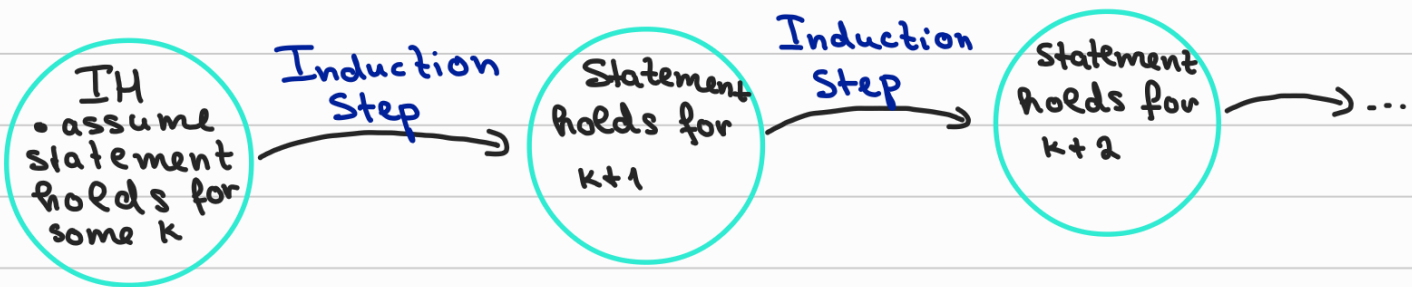
See Example 3

* Sometimes it is NOT enough to show in the base case that the statement holds for 1 number. Sometimes we need to show this for several numbers depending on the induction step.

How do the Base Case, Induction Hypothesis and Induction Step fit together?

In the Induction Hypothesis we assume that the statement holds for some $k \in \mathbb{N}$. Using this assumption in the Induction Step we show that the statement holds for $k+1$. Since we have chosen k arbitrarily and we have shown that if the statement holds for k , then it holds for $k+1$.

$\Rightarrow$ since we know that the statement holds for $k+1$ it automatically follows that the statement holds for $k+2$ and so on. This allows us to show that the statement holds for all natural numbers



It looks good, however in the moment the correctness of our proof depends on whether there actually exists such k that the statement holds for it. The goal of the Base Case is exactly to show that there is a number for which the statement holds.

Examples

Ex 1 (From the lecture)

Prove by induction that $S_n = 2^n - 1 \quad \forall n \in \mathbb{N}$, where
 $S_n = 1 + 2 + 4 + \dots + 2^{n-1}$

Solution

Base Case: We will show that the statement holds for $n = 1$

$$S_1 = 1 \quad \text{and} \quad 2^n - 1 = 2^1 - 1 = 2 - 1 = 1$$

$$\Rightarrow S_1 = 2^1 - 1$$

Hypothesis: We assume that the statement holds for $n = k, k \in \mathbb{N}$

$$\Rightarrow S_k = 1 + 2 + 4 + \dots + 2^{k-1} = 2^k - 1$$

Induction Step: We will prove that the statement holds for $n = k+1, k \in \mathbb{N}$.

$$\text{We want to prove: } S_{k+1} = 1 + 2 + 4 + \dots + 2^{k-1} + 2^k = 2^{k+1} - 1$$

$$S_{k+1} = 1 + 2 + 4 + \dots + 2^{k-1} + 2^k = \underbrace{(1 + 2 + 4 + \dots + 2^{k-1})}_{S_k} + 2^k$$

$$\Rightarrow S_{k+1} = S_k + 2^k$$

From the Induction Hypothesis we know that:

$$S_k = 2^k - 1$$

$$\Rightarrow S_{k+1} = S_k + 2^k = 2^k - 1 + 2^k = 2 \cdot 2^k - 1 = 2^{k+1} - 1$$

$$\Rightarrow S_{k+1} = 2^{k+1} - 1$$

$\Rightarrow$ We have shown that the statement holds for $n=k+1$

$\Rightarrow$ the statement holds for $\forall n \in \mathbb{N} \rightarrow$ don't forget the conclusion

Exercise 0.2 Induction. (From Exercise Sheet 0)

(a) Prove by mathematical induction that for any positive integer n ,

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}.$$

In your solution, you should address the base case, the induction hypothesis and the induction step.

(b) **(This subtask is from August 2019 exam).** Let $T : \mathbb{N} \rightarrow \mathbb{R}$ be a function that satisfies the following two conditions:

$$T(n) \geq 4 \cdot T\left(\frac{n}{2}\right) + 3n \quad \text{whenever } n \text{ is divisible by } 2;$$

$$T(1) = 4.$$

Prove by mathematical induction that

$$T(n) \geq 6n^2 - 2n$$

holds whenever n is a power of 2, i.e., $n = 2^k$ with $k \in \mathbb{N}_0$. In your solution, you should address the base case, the induction hypothesis and the induction step.

Solution

(a) Base Case: We will show that the statement holds for

$$n = 1$$

$$\frac{n(n+1)}{2} = \frac{1(1+1)}{2} = \frac{1 \cdot 2}{2} = 1 \quad \checkmark$$

Hypothesis: We assume that the statement holds for $n=k$,
 $k \in \mathbb{N}$

$$\Rightarrow 1 + 2 + \dots + k = \frac{k(k+1)}{2}$$

Induction Step: We will show that the statement holds for
 $n=k+1$.

$$\text{We want to prove } 1 + 2 + \dots + k + (k+1) = \frac{(k+1)(k+1+1)}{2} = \frac{(k+1)(k+2)}{2}$$

From the Induction Hypothesis we know that:

$$1 + 2 + \dots + k = \frac{k(k+1)}{2}$$

$$\Rightarrow 1 + 2 + \dots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+2)(k+1)}{2}$$

$$\Rightarrow 1 + 2 + \dots + k + (k+1) = \frac{(k+1)(k+2)}{2}$$

$\Rightarrow$ We have shown that the statement holds for $n = k+1$

$\Rightarrow$ We have shown that the statement holds for $\forall n \in \mathbb{N}$

(b) We will prove that the statement holds for every n , such that $n = 2^k$, $k \in \mathbb{N}_0$

Base Case: We will prove that the statement holds for $k=0$

$$\Rightarrow n = 2^0 = 1$$

$$\Rightarrow T(n) = T(1) = 4 \quad \text{and} \quad 6n^2 - 2n = 6 \cdot 1^2 - 2 \cdot 1 = 6 - 2 = 4$$

$$\Rightarrow T(1) = 4 \geq 4$$

$$\Rightarrow T(1) \geq 6 \cdot 1^2 - 2 \cdot 1$$

Hypothesis: We assume that the statement holds for $n = m = 2^l$, $m \in \mathbb{N}$, $l \in \mathbb{N}_0$

$$\Rightarrow T(m) \geq 6m^2 - 2m$$

Induction Step: We will prove that the statement holds for $n = 2m = 2^{l+1}$

We want to show that $T(2m) \geq 6(2m)^2 - 2(2m)$

We know that $2m$ is even ($m \in \mathbb{N}$)

$$\Rightarrow T(2m) \geq 4 \cdot T(m) + 3 \cdot 2m$$

$$\geq 4 \cdot (6m^2 - 2m) + 3 \cdot 2m \quad (\text{from I.H. } T(m) \geq 6m^2 - 2m)$$

$$\geq 24m^2 - 8m + 6m$$

$$= 24m^2 - 2m$$

$$\geq 24m^2 - 4m \quad (m \in \mathbb{N})$$

$$= 6 \cdot (2m)^2 - 2(2m)$$

$$\Rightarrow T(2m) \geq 6 \cdot (2m)^2 - 2(2m)$$

$\Rightarrow$ We have shown that the statement holds for $n = 2^{l+1}$

$\Rightarrow$ the statement holds for $\forall n = 2^k$, $k \in \mathbb{N}$

Ex 3 A&D Exercise Sheet 1 HS24

(a) Prove by mathematical induction that for every integer $n \geq 0$
 $1 + 3 + 5 + \dots + (2n+1) = (n+1)^2$

(b) Consider the recursive formula defined by $a_1 = 2$ and

$a_{n+1} = 6a_n - 2$ for $n \geq 1$. Determine the smallest positive integer m such that $a_m > 2^{2^m}$. Then, prove by induction that $a_n \geq 2^{2^n}$ for all integers $n \geq m$.

Solution

Base case: We will prove that the statement holds for $n=0$
 $1 = (0+1)^2$

Hypothesis: We assume that the statement holds for $n=k$,
 $k \in \mathbb{N}_0$

$$\Rightarrow 1+3+5+\dots+(2k+1) = (k+1)^2$$

Induction Step: We will show that the statement holds for $n=k+1$

$$\begin{aligned} & \underbrace{1+3+5+\dots+(2k+1)}_{(k+1)^2} + (2(k+1)+1) = \\ & (k+1)^2 + (2k+2+1) = \quad (\text{from I.H. } 1+3+5+\dots+(2k+1) = (k+1)^2) \\ & k^2 + 2k + 1 + 2k + 3 = \\ & k^2 + 4k + 4 = \\ & (k+2)^2 = \\ & ((k+1)+1)^2 \end{aligned}$$

$$\Rightarrow 1+3+5+\dots+(2k+1) + (2(k+1)+1) = ((k+1)+1)^2$$

$\rightarrow$ We have shown that the statement holds for $n=k+1$

$\Rightarrow$ We have shown that the statement holds for $\forall n \in \mathbb{N}_0$

(B) First we will determine the smallest m such that $a_m > 2^{2^m}$

$$a_1 = 2 \text{ and } 2^{2 \cdot 1} = 4 \Rightarrow a_1 < 2^{2 \cdot 1}$$

$$a_2 = 6 \cdot a_1 - 2 = 6 \cdot 2 - 2 = 10 \text{ and } 2^{2 \cdot 2} = 16 \Rightarrow a_2 < 2^{2 \cdot 2}$$

$$a_3 = 6 \cdot a_2 - 2 = 6 \cdot 10 - 2 = 58 \text{ and } 2^{2 \cdot 3} = 64 \Rightarrow a_3 < 2^{2 \cdot 3}$$

$$a_4 = 6 \cdot a_3 - 2 = 348 - 2 = 346 \text{ and } 2^{2 \cdot 4} = 256 \Rightarrow a_4 > 256$$

$\Rightarrow$ the smallest m , such that $a_m > 2^{2^m}$ is $m=4$

Now we will prove by induction that $a_n > 2^{2^n} \forall n \in \mathbb{N}$ and $n \geq m$

Base Case: $n=4$

Hypothesis: We assume the statement holds for $n=k, k \in \mathbb{N}, k \geq 4$

$$\Rightarrow a_k \geq 2^{2k}$$

Induction Step: We will show that the statement holds for $n = k+1$, $k \in \mathbb{N}$, $k \geq m$

We want to prove that $a_{k+1} \geq 2^{2(k+1)}$

$$a_{k+1} = 6a_k - 2$$

$$\geq 6 \cdot 2^{2k} - 2 \quad (\text{from I.H. } a_k \geq 2^{2k})$$

$$= 2 \cdot 2^{2k} + 4 \cdot 2^{2k} - 2$$

$$= 2^{2k+2} + 2^{2k+1} - 2$$

$$\geq 2^{2k+2} \quad (\text{since } k \geq 4 \quad 2^{2k+1} - 2 > 0)$$

$$= 2^{2(k+1)}$$

$$\Rightarrow a_{k+1} \geq 2^{2(k+1)}$$

$\Rightarrow$ We have shown that the statement holds for $n = k+1$

$\Rightarrow$ We have shown that the statement holds for $\forall n \in \mathbb{N}$, $n \geq 4$

Strong induction

Strong induction has the same steps as the normal induction, however in the Induction Hypothesis instead of assuming that the statement holds for some $k \in \mathbb{N}$, we assume that the statement holds for all numbers from 1 to k included.

Examples

Ex 4 We have unlimited number of coins worth 4 and unlimited number of coins worth 5. Show that any sum n , $n \in \mathbb{N}$, $n \geq 12$ can be paid using these coins.

Solution

Base Case: $n = 12$

$$12 = 3 \text{ coins worth } 4$$

$$n = 13$$

$$13 = 1 \text{ coin worth } 5 + 2 \text{ coins worth } 4$$

$$n = 14$$

$$14 = 2 \text{ coins worth } 5 + 1 \text{ coin worth } 4$$

$$n = 15$$

$$15 = 3 \text{ coins worth } 5$$

Hypothesis: We assume that all sums from 12 to $n = k$ can be paid with the coins.

Induction Step: We will show that the statement holds for $n = k+1$. Since in the Base case we already saw that all sums from 12 to 15 included can be paid with the coins we consider only the case when $k \geq 15$ and $k+1 \geq 16$

$$\Rightarrow (k+1) - 4 \geq 12 \quad \text{and} \quad (k+1) - 4 \leq k$$

$\Rightarrow$ from the Induction Hypothesis we know that all sums from 12 to k can be paid with the coins including $(k+1) - 4$

$\Rightarrow$ Since $(k+1) - 4$ can be paid with the coins, by adding 1 coin worth 4 we can also pay $k+1$

$\Rightarrow$ We have shown that the statement holds for $n = k+1$, $k \geq 12$

$\Rightarrow$ We have shown that the statement holds for $\forall n \in \mathbb{N}, n \geq 12$

Ex 5 Prove that all horses are the same color.

It is enough to prove that in every hurdle of n horses all horses are the same color for $\forall n \in \mathbb{N}$.

Solution

Base case: $n = 1$

Since the hurdle contains only 1 horse, all horses in the hurdle are the same color.

Hypothesis: We assume that the statement holds for $n = k$
 $\Rightarrow$ in all hurdles of size k all horses are the same color.

Induction Step: We will prove that the statement holds for $n = k+1$.

Let us remove 1 horse from the hurdle. We call this horse H_1 .

$\Rightarrow$ without H_1 there are k horses left in the hurdle and from the induction hypothesis follows that all k horses

have the same color. Let that color be Brown (without loss of generality). Now we remove 1 horse from the k horses. We call it H_2 . We know that H_2 is Brown. Now we add H_1 to the $k-1$ horses and we get a herd of k horses. From the induction hypothesis we know that all k horses have the same color and since all $k-1$ horses that are not H_1 are Brown, H_1 is also Brown.

$\Rightarrow$ All $k+1$ horses are Brown

$\Rightarrow$ We have shown the statement holds for $n=k$

$\Rightarrow$ We have shown the statement holds for $\forall n \in \mathbb{N}$

Where is the problem?

The key part of the induction step are the $k-1$ horses that are the intersection of the two herds of size k that include H_1 and H_2 respectively. When $k-1 \geq 1$, the induction step is correct, however when $k-1=0$, the horses H_1 and H_2 can be different colors.

$\Rightarrow$ in order our induction to be correct we need to choose $k \geq 2$, however if $k \geq 2$ we need to prove that the statement holds for $k \geq 2$ in the Base Case, but we cannot do that.

Limit of a Function

Informal Explanation A limit is the value that a function "approaches" as the argument gets closer and closer to a certain point. It describes the behaviour of the function when the argument gets near that point, rather than the function's value at the point itself.

$\lim_{x \rightarrow y} f(x)$, where f is the function, x is the argument, y is the point to which the argument goes closer and closer to.

Useful Rules Let $f(x), g(x)$ be two functions, such that $\lim_{x \rightarrow y} f(x) = c$ $\lim_{x \rightarrow y} g(x) = d$, where $c, d \in \mathbb{R}$ (this means $c, d \neq \infty$)

$$1) \lim_{x \rightarrow y} f(x) + g(x) = \lim_{x \rightarrow y} f(x) + \lim_{x \rightarrow y} g(x) = c + d$$

$$2) \lim_{x \rightarrow y} f(x) - g(x) = \lim_{x \rightarrow y} f(x) - \lim_{x \rightarrow y} g(x) = c - d$$

$$3) \lim_{x \rightarrow y} f(x) \cdot g(x) = \lim_{x \rightarrow y} f(x) \cdot \lim_{x \rightarrow y} g(x) = c \cdot d$$

$$4) \text{ if } \lim_{x \rightarrow y} g(x) = d \neq 0$$

$$\lim_{x \rightarrow y} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow y} f(x)}{\lim_{x \rightarrow y} g(x)} = \frac{c}{d}$$

! Those rules don't always hold if $c = \infty$ or $d = \infty$.

! Rule 4) doesn't mean that $\lim_{x \rightarrow y} \frac{f(x)}{g(x)}$ doesn't exist when

$\lim_{x \rightarrow y} g(x) = 0$. It only means that it cannot be calculated using rule 4.

Asymptotic Growth

When we estimate the number of elementary operations executed by algorithms, it is often useful to ignore smaller order terms, and instead focus on the asymptotic growth defined below. We denote by $\mathbb{R}^+$ the set of all (strictly) positive real numbers and by $\mathbb{R}_0^+$ the set of nonnegative real numbers.

Definition 1. Let $f, g : \mathbb{N} \rightarrow \mathbb{R}^+$ be two functions. We say that f grows asymptotically faster than g if $\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = 0$.

This definition is also valid for functions defined on $\mathbb{R}^+$ instead of $\mathbb{N}$. In general, $\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)}$ is the same as $\lim_{x \rightarrow \infty} \frac{g(x)}{f(x)}$ if the second limit exists.

Good to know

$$1) \lim_{n \rightarrow \infty} \frac{\log n}{n} = 0 \Rightarrow n \text{ grows asymptotically faster than } \log n$$

$$2) \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0 \Rightarrow n^n \text{ grows asymptotically faster than } \log n$$

$$3) \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad 4) \lim_{n \rightarrow \infty} q^n = 0 \text{ for } q \in \mathbb{R} \text{ and } |q| < 1$$

$$5) \lim_{n \rightarrow \infty} \frac{n^c}{k^n} = 0, \text{ where } c \in \mathbb{R}, k \in \mathbb{R}^+ \text{ and } |k| > 1 \text{ are constants}$$

Exercises

Exercise 0.2 Comparison of functions part 1. Exercise Sheet 0 HS 25

Show that

- (a) $f(n) := n \log n$ grows asymptotically faster than $g(n) := n$.
- (b) $f(n) := n^3$ grows asymptotically faster than $g(n) := 10n^2 + 100n + 1000$.
- (c) $f(n) := 3^n$ grows asymptotically faster than $g(n) := 2^n$.

Solution:

In order to show that $f(n)$ grows asymptotically faster than $g(n)$, we will show that $\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = 0$

$$a) \lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = \lim_{n \rightarrow \infty} \frac{n}{n \log n} = \lim_{n \rightarrow \infty} \frac{1}{\log n} = 0$$

$$b) \lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = \lim_{n \rightarrow \infty} \frac{10n^2 + 100n + 1000}{n^3} = \lim_{n \rightarrow \infty} \left(\frac{10}{n} + \frac{100}{n^2} + \frac{1000}{n^3} \right) = 0$$

$$c) \lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = \lim_{n \rightarrow \infty} \frac{2^n}{3^n} = \lim_{n \rightarrow \infty} \left(\frac{2}{3} \right)^n = 0$$

Theorem 1 (L'Hôpital's rule). Assume that functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ and $g : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ are differentiable, $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = \infty$ and for all $x \in \mathbb{R}^+$, $g'(x) \neq 0$. If $\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = C \in \mathbb{R}_0^+$ or $\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \infty$, then

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}.$$

Useful Rules

Let f and g be differentiable functions

$$1) (f(x) + g(x))' = f'(x) + g'(x)$$

$$2) (f(x) - g(x))' = f'(x) - g'(x)$$

$$3) (f(x) \cdot g(x))' = f'(x)g(x) + f(x)g'(x)$$

$$4) \left(\frac{f(x)}{g(x)} \right)' = \frac{f'(x)g(x) - g'(x)f(x)}{g^2(x)}, \text{ when } g(x) \neq 0$$

$$5) (c \cdot f(x))' = c \cdot f'(x), \text{ where } c \in \mathbb{R} \text{ is a constant}$$

$$6) (f(x) + c)' = f'(x), \text{ where } c \in \mathbb{R} \text{ is a constant}$$

Table of derivatives

function	first derivative
$f(x) = c, c \in \mathbb{R}$	$f'(x) = 0$
$f(x) = x$	$f'(x) = 1$
$f(x) = x^n, n \in \mathbb{R}$	$f'(x) = n \cdot x^{n-1}$
$f(x) = e^x$	$f'(x) = e^x$
$f(x) = \ln x$	$f'(x) = \frac{1}{x}$
$f(x) = \log_a(x), a \in \mathbb{R}^+$	$f'(x) = \frac{1}{x \ln(a)}$

Exercise 0.3 Comparison of functions part 2.

Show that

- (a) $f(n) := n^{1.01}$ grows asymptotically faster than $g(n) := n \ln n$.
- (b) $f(n) := e^n$ grows asymptotically faster than $g(n) := n$.
- (c) $f(n) := e^n$ grows asymptotically faster than $g(n) := n^2$.
- (d)* $f(n) := 1.01^n$ grows asymptotically faster than $g(n) := n^{100}$.
- (e) $f(n) := \log_2 n$ grows asymptotically faster than $g(n) := \log_2 \log_2 n$.
- (f) $f(n) := 2^{\sqrt{\log_2 n}}$ grows asymptotically faster than $g(n) := \log_2^{100} n$.
- (g) $f(n) := n^{0.01}$ grows asymptotically faster than $g(n) := 2^{\sqrt{\log_2 n}}$.

Solution:

In order to show that $f(n)$ grows asymptotically faster than $g(n)$ we will show that $\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = 0$

$$\begin{aligned} \text{a) } \lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} &= \lim_{n \rightarrow \infty} \frac{n \ln n}{n^{1.01}} = \lim_{n \rightarrow \infty} \frac{\ln n}{n^{0.01}} \stackrel{\text{Thm 1}}{=} \lim_{n \rightarrow \infty} \frac{(\ln n)'}{(n^{0.01})'} = \\ &= \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{0.01 \cdot n^{-0.99}} = \lim_{n \rightarrow \infty} \frac{n^{0.99}}{0.01 \cdot n} \stackrel{\text{Thm 1}}{=} \lim_{n \rightarrow \infty} \frac{100}{n^{0.01}} = 0 \end{aligned}$$

$$\text{b) } \lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = \lim_{n \rightarrow \infty} \frac{n}{e^n} \stackrel{\text{Thm 1}}{=} \lim_{n \rightarrow \infty} \frac{n'}{(e^n)'} = \lim_{n \rightarrow \infty} \frac{1}{e^n} = 0$$

$$\begin{aligned} \text{c) } \lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} &= \lim_{n \rightarrow \infty} \frac{n^2}{e^n} \stackrel{\text{Thm 1}}{=} \lim_{n \rightarrow \infty} \frac{(n^2)'}{(e^n)'} = \lim_{n \rightarrow \infty} \frac{2n}{e^n} \stackrel{\text{Thm 1}}{=} \lim_{n \rightarrow \infty} \frac{(2n)'}{(e^n)'} = \\ &= \lim_{n \rightarrow \infty} \frac{2}{e^n} = 0 \end{aligned}$$

$$d) \lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = \lim_{n \rightarrow \infty} \frac{n^{100}}{1.01^n} = \lim_{n \rightarrow \infty} \frac{e^{100 \ln n}}{e^{n \ln 1.01}} =$$

$$\lim_{n \rightarrow \infty} e^{(100 \ln n - n \ln(1.01))} = e^{\lim_{n \rightarrow \infty} 100 \ln n - n \ln(1.01)}$$

We have

$$\lim_{n \rightarrow \infty} 100 \ln n - n \ln(1.01) = \lim_{n \rightarrow \infty} n \left(100 \frac{\ln n}{n} - \ln(1.01) \right)$$

$$= \lim_{n \rightarrow \infty} n \cdot \lim_{n \rightarrow \infty} \left(100 \frac{\ln n}{n} - \ln(1.01) \right)$$

$$= \lim_{n \rightarrow \infty} n \cdot (-\ln(1.01)) = -\infty$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = \lim_{n \rightarrow \infty} e^{(100 \ln n - n \ln(1.01))} = e^{\lim_{n \rightarrow \infty} 100 \ln n - n \ln(1.01)}$$

$$= e^{-\infty} = 0$$

$$e) \lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = \lim_{n \rightarrow \infty} \frac{\log_2 \log_2 n}{\log_2 n}$$

$$\text{Let } y = \log_2 n$$

$$\text{Since } n \rightarrow \infty \text{ and } y = \log_2 n \Rightarrow y \rightarrow \infty$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{\log_2 \log_2 n}{\log_2 n} = \lim_{y \rightarrow \infty} \frac{\log_2 y}{y}$$

$$\log_2 y = \frac{\ln y}{\ln 2}$$

* Change of basis $\log_a x = \frac{\log_e x}{\log_e a}$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{\log_2 \log_2 n}{\log_2 n} = \lim_{y \rightarrow \infty} \frac{\log_2 y}{y} = \lim_{y \rightarrow \infty} \frac{\frac{\ln y}{\ln 2}}{y} = \lim_{y \rightarrow \infty} \frac{\ln y}{\ln 2 \cdot y} =$$

$$\stackrel{\uparrow \text{Thm 1}}{=} \lim_{y \rightarrow \infty} \frac{(\ln y)'}{(\ln 2 \cdot y)'} = \lim_{y \rightarrow \infty} \frac{\frac{1}{y}}{\ln 2} = \lim_{y \rightarrow \infty} \frac{1}{y \ln 2} = 0$$

Alternatively

$$\lim_{n \rightarrow \infty} \frac{\log_2 \log_2 n}{\log_2 n} = \lim_{y \rightarrow \infty} \frac{\log_2 y}{y} \stackrel{\uparrow \text{Thm 1}}{=} \lim_{y \rightarrow \infty} \frac{(\log_2 y)'}{y'} = \lim_{y \rightarrow \infty} \frac{\frac{1}{y \ln 2}}{1}$$

$$= \lim_{y \rightarrow \infty} \frac{1}{y \ln 2} = 0$$

$$\begin{aligned} f) \lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} &= \lim_{n \rightarrow \infty} \frac{\log_2^{100} n}{2^{\sqrt{\log_2 n}}} = \lim_{n \rightarrow \infty} \frac{2^{\log_2 \log_2^{100} n}}{2^{\sqrt{\log_2 n}}} = \lim_{n \rightarrow \infty} \frac{2^{100 \log_2 \log_2 n}}{2^{\sqrt{\log_2 n}}} \\ &= \lim_{n \rightarrow \infty} 2^{100 \log_2 \log_2 n - \sqrt{\log_2 n}} = 2^{\lim_{n \rightarrow \infty} 100 \log_2 \log_2 n - \sqrt{\log_2 n}} \end{aligned}$$

$$\lim_{n \rightarrow \infty} 100 \log_2 \log_2 n - \sqrt{\log_2 n} = \lim_{n \rightarrow \infty} \sqrt{\log_2 n} \left(100 \frac{\log_2 \log_2 n}{\sqrt{\log_2 n}} - 1 \right) =$$

$$\lim_{n \rightarrow \infty} \sqrt{\log_2 n} \cdot \lim_{n \rightarrow \infty} \left(100 \frac{\log_2 \log_2 n}{\sqrt{\log_2 n}} - 1 \right) = \lim_{n \rightarrow \infty} \sqrt{\log_2 n} \cdot (-1) = -\infty$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = 2^{\lim_{n \rightarrow \infty} 100 \log_2 \log_2 n - \sqrt{\log_2 n}} = 2^{-\infty} = 0$$

$$\begin{aligned} g) \lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} &= \lim_{n \rightarrow \infty} \frac{2^{\sqrt{\log_2 n}}}{n^{0,01}} = \lim_{n \rightarrow \infty} \frac{2^{\sqrt{\log_2 n}}}{2^{0,01 \cdot \log_2 n}} = \lim_{n \rightarrow \infty} 2^{\sqrt{\log_2 n} - 0,01 \log_2 n} \\ &= 2^{\lim_{n \rightarrow \infty} \sqrt{\log_2 n} - 0,01 \log_2 n} \end{aligned}$$

$$\lim_{n \rightarrow \infty} \sqrt{\log_2 n} - 0,01 \log_2 n = \lim_{n \rightarrow \infty} \log_2 n \left(\frac{1}{\sqrt{\log_2 n}} - 0,01 \right) =$$

$$\lim_{n \rightarrow \infty} \log_2 n \cdot (-0,01) = -\infty$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{2^{\sqrt{\log_2 n}}}{n^{0,01}} = 2^{\lim_{n \rightarrow \infty} \sqrt{\log_2 n} - 0,01 \log_2 n} = 2^{-\infty} = 0$$

Exercise 0.4 Simplifying expressions.

Simplify the following expressions as much as possible without changing their asymptotic growth rates.

Concretely, for each expression $f(n)$ in the following list, find an expression $g(n)$ that is as simple as possible and that satisfies $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} \in \mathbb{R}^+$.

(a) $f(n) := 5n^3 + 40n^2 + 100$ n^3

(b) $f(n) := 5n + \ln n + 2n^3 + \frac{1}{n}$ n^3

(c) $f(n) := n \ln n - 2n + 3n^2$ n^2

(d) $f(n) := 23n + 4n \log_5 n^6 + 78\sqrt{n} - 9$ $n \ln n$

(e) $f(n) := \log_2 \sqrt{n^5} + \sqrt{\log_2 n^5}$ $\ln n$

(f)* $f(n) := 2n^3 + \left((\sqrt[4]{n})^{\log_5 \log_6 n} \right) + (\sqrt[7]{n})^{\log_8 \log_9 n}$